Benjamin Menetrier

Birth date:	14/05/1987, French national.
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Positions

- **May 2023 - present**: Researcher at the Norwegian Meteorological Institute (Oslo, Norway).
- **October 2018 - April 2023**: Research scientist at IRIT (Toulouse), subcontracting for the JCSDA (Bouder, CO). Main developer of the SABER library in the JEDI project.
- **April 2016 - September 2018**: Research scientist at CNRM (Toulouse).
- **October 2015 - March 2016**: Research engineer at CERFACS (Toulouse).
- **Septembre 2014 - September 2015**: Project Scientist I at NCAR (Boulder, CO).

Education

- **2011-2014**: PhD at CNRM (Toulouse): "Using ensemble data assimilation for the modelling of flow-dependent background error covariances at convective scale", Léopold Escande award 2014.
- **2011**: ENS (Paris) graduation, geosciences specialty.
- **2010**: Master “Ocean, Atmosphere, Climate and Remote Sensing” at UPMC (Paris).

Skills

- Language: fluent English, French mother tongue, beginner in Swedish.
- Scientific computing expertise with Fortran (15+ years), C++ and Python (10+ years).
- Developer in large, complex and distributed-memory codes of data assimilation for operational weather forecasting (IFS/ARPEGE-AROME, WRF-DA, GSI, NEMOVAR, OOPS, JEDI framework).
- International collaborative code development with high-quality standards, using git, issue tracking, code reviews, automated testing.
- Experienced Linux / bash user, strong experience as a HPC user on many platforms.

Teaching

- **2016 - 2025**: training about ensemble-variational methods for data assimilation at CERFACS (Toulouse).
- **2024**: lecture and practical training for developers on the OOPS system (ACCORD consortium).
- **2019 - 2023**: lectures and practical training for the “JEDI academies” organised by the JCSDA.

Bibliometry

- Author or co-author of 22 published papers in international peer-reviewed journals.
- Cited 476 times, H-index 13 [Web of Science, 23/10/2026].
- ORCID account: 0000-0002-2593-3398 / Web of Science account: ABG-7190-2021

Selected recent papers

- Ménétrier, B. (2026): The Normalized Interpolated Convolution from an Adaptive Subgrid (NICAS) method, *Geosci. Model Dev.*, 19, 4497-4511.
- Abdi-Oskouei *et al.* (2026): Capability Demonstration of a JEDI-Based System for TEMPO Assimilation: System Description and Evaluation, *Journal of Advances in Modeling Earth Systems*, 18, e2025MS005482.
- Lussana *et al.* (2025). Ensemble-based statistical interpolation of atmospheric variables near the surface *Quarterly Journal of the Royal Meteorological Society*, 151(772).
- Jung *et al.* (2024): Three-dimensional variational assimilation with a multivariate background error covariance for the Model for Prediction Across Scales - Atmosphere with the Joint Effort for Data assimilation Integration (JEDI-MPAS 2.0.0-beta), *Geosci. Model Dev.*, 17, 3879-3895.
- Guerette *et al.* (2024): Data assimilation for the Model for Prediction Across Scales - Atmosphere with the Joint Effort for Data assimilation Integration (JEDI-MPAS 2.0.0-beta): ensemble of 3D ensemble-variational (En-3DEnVar) assimilations. *Geosci. Model Dev.*, 16, 7123-7142.